<Add your name and student ID (SID) to the footer. Add your answers below. Include any relevant output from Excel with your answer. To receive full marks for probability questions, include your working as well as the final answer e.g. a copy of the formula used in Excel. Some questions may be easier to handwrite. In those cases scan in your answer and include your scan at the relevant point.>

Background

Domain.com.au is a source of 'big data'. The site lists properties that have recently been sold. On 16/9/2025 Microsoft copilot was used to scrape the data of house and apartment sold between 30/6/2025 and 15/9/2025 in Orange. An example page used was: https://www.domain.com.au/sold-listings/orange-nsw-2800/house/?excludepricewithheld=1

The processed data is available from "20250916OrangeSales.xlsx"

Imagine you are a real estate agent in Orange. You want to use the data to inform the advice you give to clients and to create compelling presentations that attract new clients.

All statistics, tables and charts should be generated in Excel.

You are allowed to use generative AI to assist with Excel; interpretation should be done by you. Some questions will indicate when generative AI is not allowed. Anytime you use generative AI, include the summary of prompts and responses (ask the AI to create the summary for you).

Do not submit the transformed excel file. However, you will need to give screenshots to show you have used Excel successfully.

Question 1: Initial Investigation of the data set and variables (12 marks)

a. **State** the number of variables in the data set. (1 mark)

11 variables found in the excel spreadsheet.

b. **State** the number of records in the data set. (1 mark)

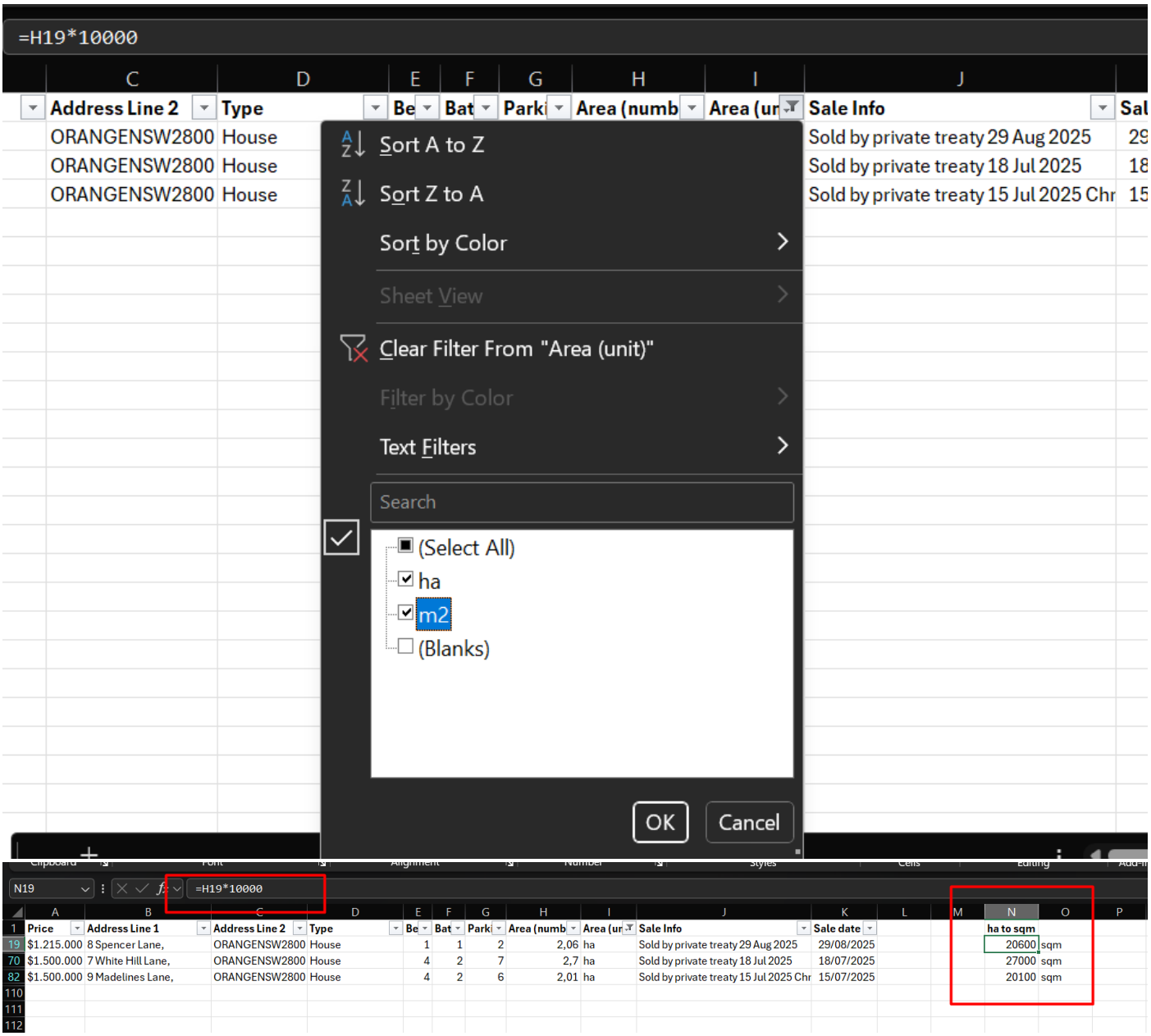
108 records found in the data set

c. Most properties list the ***area*** of the property. Note: not all properties use the same ***unit*** of measurement.

List all the units used.
Convert all entries into square metres and **show** the entries you have converted. (1+2 marks)

Tips: Using the data menu, turn on filters. Filters can be used to sort or filter the data. Via sorting or filtering, find all properties that have an area not recorded in square metres and then convert them. There are only a few, so you may convert them manually. Alternatively, make a new column. Take a screenshot of the rows you modify (showing the address line and converted area).

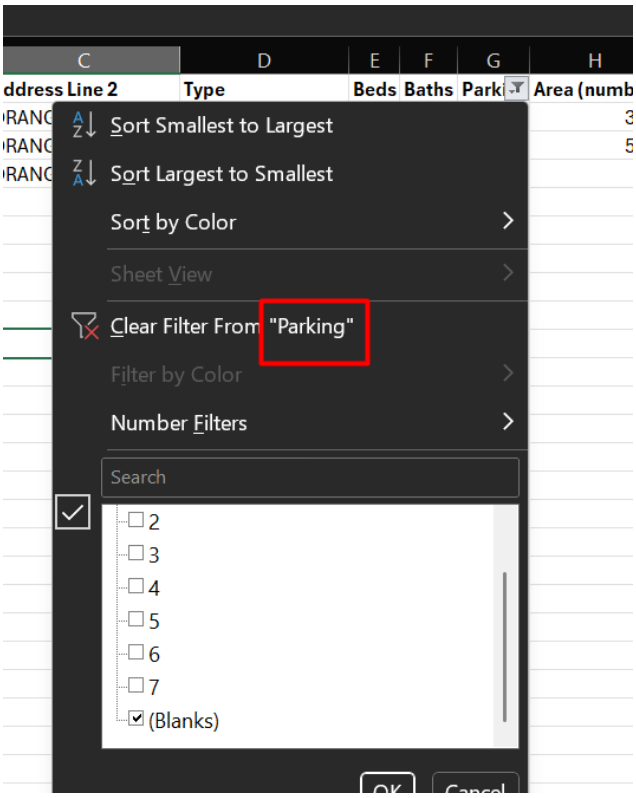
There are only 2 units used in the area, ha and sqm.



d. Some properties do not list the amount of ***parking*** (garages).

List all properties that have missing data for parking.
For each property, **give** the most likely reason the data is missing from the spreadsheet.
Replace the missing data with actual numbers (as much as possible). (1+2+1 marks)

*Tips: Filter or sort the data using the **parking** column to find the rows with missing data. Use the values in other columns to figure out why the data might be missing. This is real data, so if you are stuck do a web search of the relevant addresses for more hints. Take a screenshot of the rows you modified.*



There are missing parking values because of small properties, these commonly do not include dedicated parking spaces. Therefore, it is reasonable to change these missing values to zero parking spaces.

Price	Address Line 1	Address Line 2	Type	Beds	Baths	Park	Ar
$547.500	144 Edward Street,	ORANGENSW2800	Semi-detached	3	1	0	
$467.500	15 Algona Crescent,	ORANGENSW2800	House	3	1	0	
$768.000	40 Lister Drive,	ORANGENSW2800	House			0	

State the type of the variable ***Baths***. (1 mark)

Baths is a quantitative variable, it represents a count of bathrooms.

e. The data is semi-structured, with much of it being organised in columns, but there are unstructured parts. **Identify** the column(s) which is/are unstructured and explain what makes each unstructured. (2 marks)

While the most part of the data set is structured, there are some unstructured columns: Address Line 1, Address Line 2 and Sale Info consist of free-text that do not follow a consistent format.

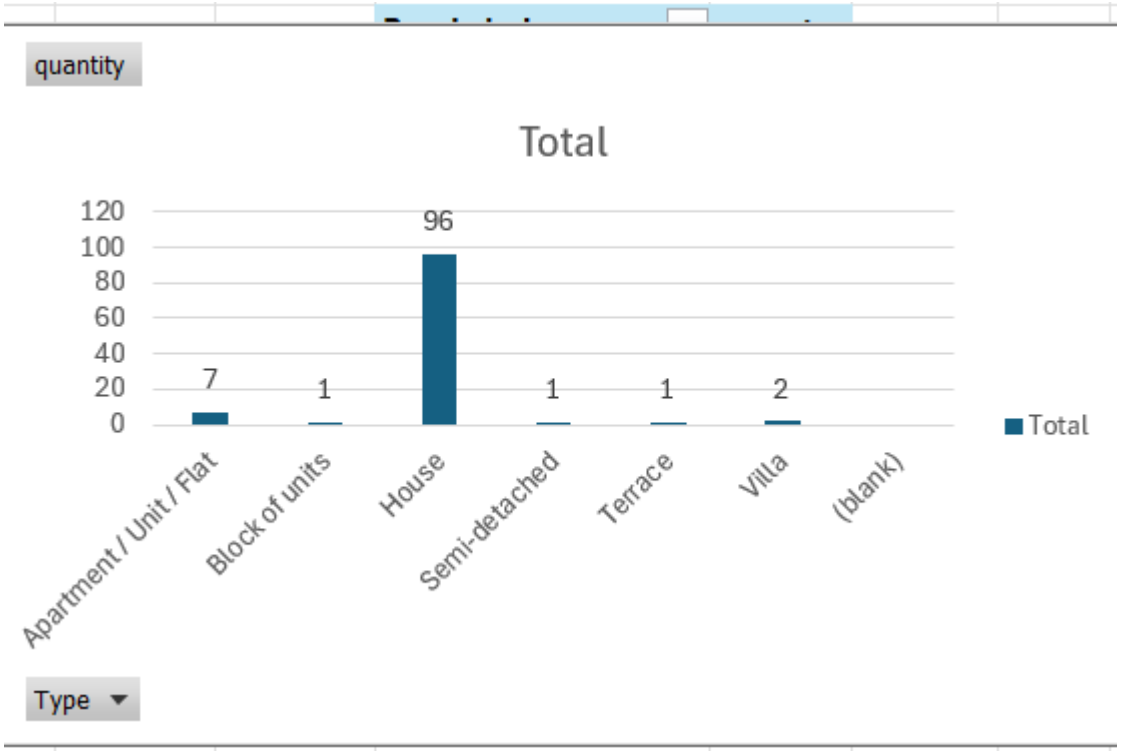
Question 2: Investigating the variable "Type" (7 marks)

a. **Generate** a pivot table of the variable ***Type***. Each type should be a different row. For each row show both the count and the percentage. (2 marks)

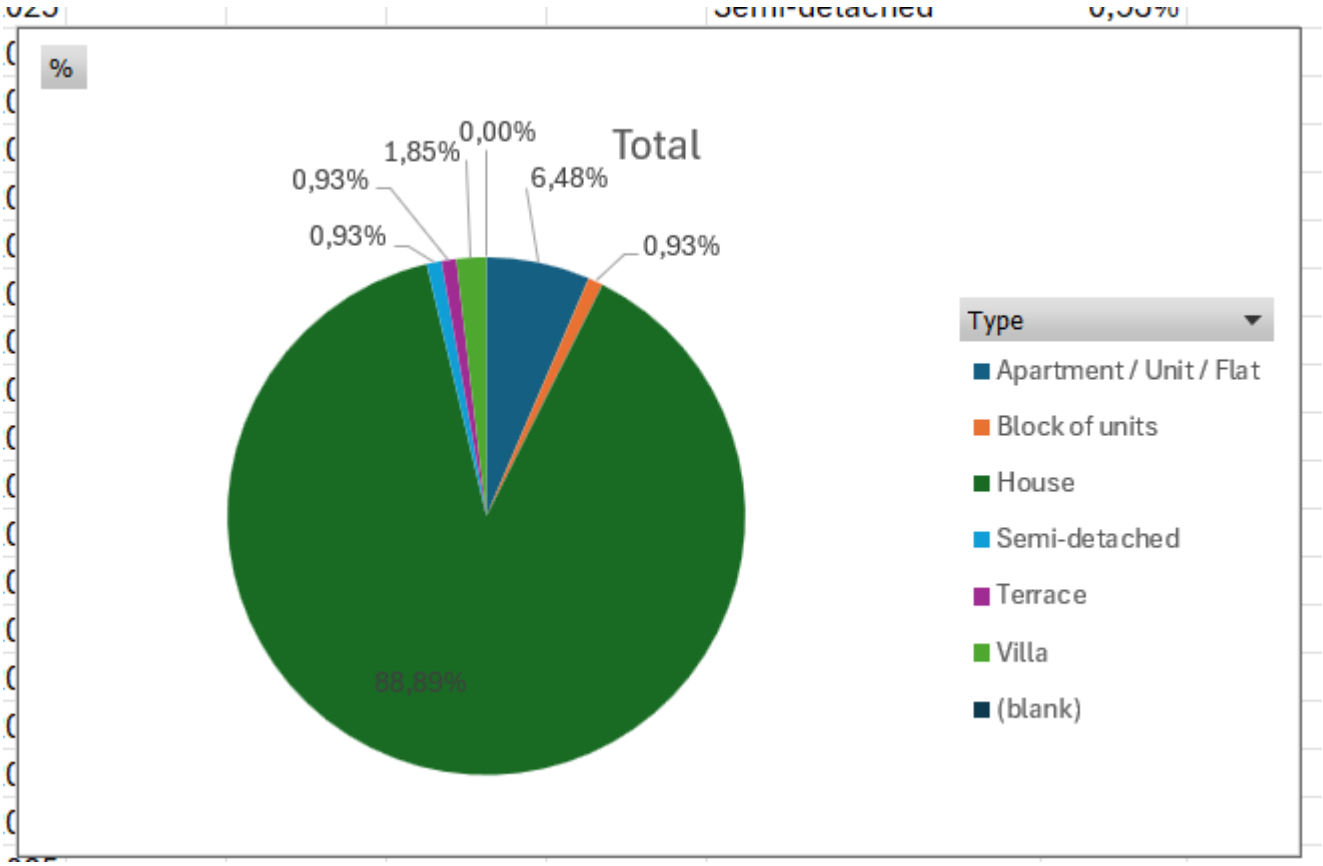
Row Labels	Count of Type	%
Apartment / Unit / Flat	7	6,48%
Block of units	1	0,93%
House	96	88,89%
Semi-detached	1	0,93%
Terrace	1	0,93%
Villa	2	1,85%
(blank)		0,00%
Grand Total	**108**	**100,00%**

b. **Create** a bar chart for the variable ***Type***. Ensure each bar displays the count of properties for that type. Also, update the title. (2 marks)

Tips: use a pivot chart based on your table. Change formatting to get the desired results.



c. **Create** a pie chart for the variable ***Type***. Ensure each segment displays the percentage for that type. Also, update the title. (2 marks)



d. To simplify the data, domain.com.au puts all ***types*** into three ***groups***: townhouses (none sold in this data set), x (includes flats, units, and blocks of units), houses (everything else).

State the percentage of properties in this snapshot that are in the 'houses' ***group***. (1 mark)

*Tips: This question can be answered by adding some numbers from your pivot table. However, to answer other questions it is recommended to create a new column called **group** to capture this information.*

Approximately **93.52%** of properties in this snapshot are classified as houses.

Row Labels	Count of Type
Apartments	6,48%
houses	93,52%
(blank)	0,00%
Grand Total	**100,00%**

Question 3: Investigating the variable "Price" (14 marks)

a. Use the Analysis ToolPak to **generate** descriptive statistics for the variable ***price***. (2 marks)

Tip: make sure you install the Analysis ToolPak Add-in first.

	price
Mean	734883
Standard Error	24695
Median	687500
Mode	600000
Standard Deviation	256638
Sample Variance	6,6E+10
Kurtosis	3,86728
Skewness	1,63998
Range	1462000
Minimum	288000
Maximum	1750000
Sum	7,9E+07
Count	108

b. **Calculate** the <u>coefficient of variation</u> (CV). (1 mark)

=Q19/Q15

K	L	M	N	O	P	Q	R	S	T
05/09/2025	Apartments					*price*			
04/09/2025	houses								
04/09/2025	houses				Mean	734883		coefficient of variation	
03/09/2025	houses				Standard Error	24695			
01/09/2025	houses				Median	687500		=Q19/Q15	
29/08/2025	houses				Mode	600000			
29/08/2025	houses		20600	sqm	Standard Deviation	256638			
29/08/2025	houses				Sample Variance	6,6E+10			
29/08/2025	houses				Kurtosis	3,86728			
28/08/2025	houses				Skewness	1,63998			

The coefficient of variation is 0.349223442

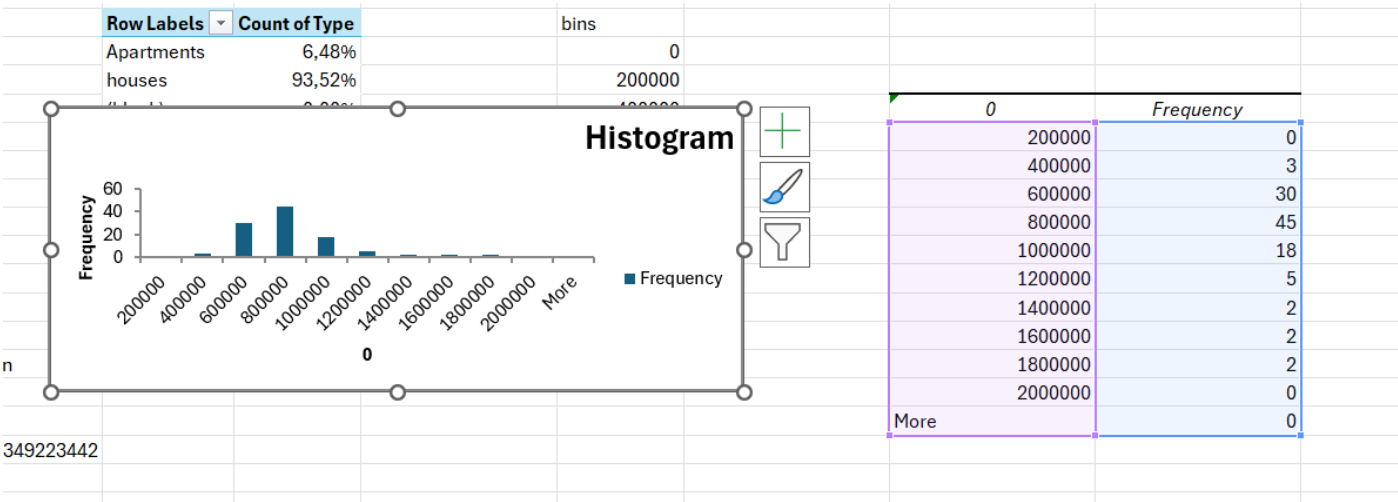
It was calculated by dividing the standard deviation by the mean (CV = standard deviation /mean)

c. Use Excel to **generate** a <u>histogram</u> of price.

Make the first bin start at $0 and make the bin width $200,000. Ensure your histogram includes all data points by using a sufficient number of bins. Format the graph appropriately, including a title

and axes that reflect the data. (4 marks)

Tips: Excel has a few ways to do histograms – all with various trade-offs (no method does everything automatically unfortunately). You could use generative AI to give you the pros and cons of the approaches. Our recommended approach is to use the Analysis ToolPak. However, some editing of the formatting (including title, axes, widening bars) is needed (see topic 4.1, example 3.2 for an illustration). You can take other approaches though. Marking: 1 mark for histogram with overall correct shape and including all data points, 1 mark for histogram starting at $0, 1 mark for bin width of $200,000, 1 mark for correct formatting (title, axes labels, no gaps between columns).



d. **State** the symmetry of the data (symmetric, left skewed or right skewed).

Justify your response using both the descriptive statistics (in two different ways) and histogram. (1 + 2 + 1 marks)

Symmetry:
Justification 1:

The skewness from descriptive statistics is positive, that confirms it is right skewness.

Justification 2:
The mean price is bigger than median price. That confirms it is right skewness

Justification 3:

The histogram shows a long tail to the right. Which means, the most property are clustered are lower price and a few at higher price. That means it is right skewness.

e. **Explain** whether the mean or the median is the best way to summarise the price (1 mark).

The median is the best way to summarise the price. While some houses/apartments raise mean prices due some extreme values, median summarises typical prices showing where are the majority of prices .

f. Use Excel to **summarise** the ***price*** just for Apartments/unit/flats. **Search** for a comparable historical figure from four years ago. **Write** a brief statement that a real estate agent could use to recruit new clients. (1 + 1 + 1 marks).

Tips: Apartments/unit/flats is one of the values for ***type***. *Use the summary statistic from Q3e. For the first part, you need to filter (or sort) then summarise.*

Answer here:

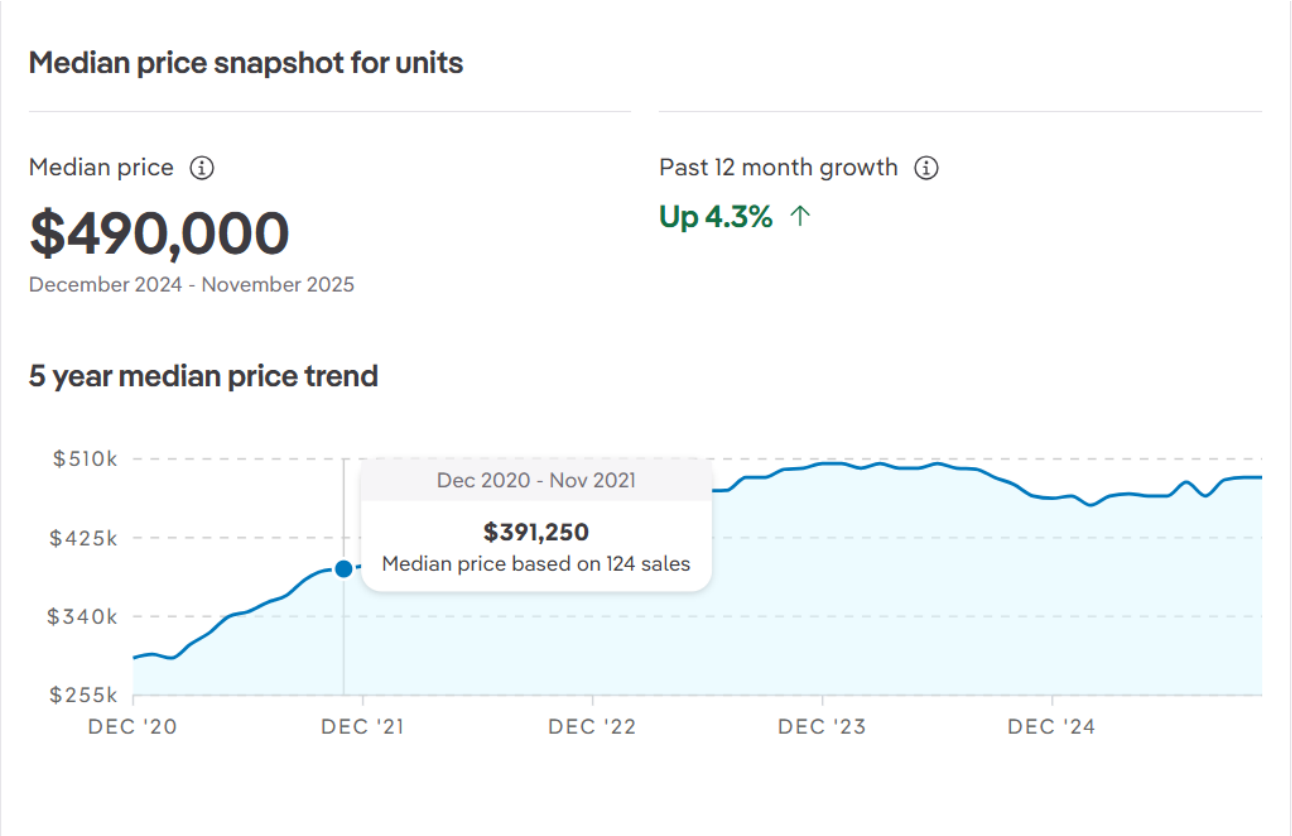
Summary of price: $687.500

O113 =MEDIAN(A8:A73)

Price	Address Line 1	Address Line 2	Type	Beds	Baths	Parking	Area (number)	Area (unit)	Sale Info	Sale date	group	ha to sqm	
$380.000	4/45-49 Frost Street,	ORANGENSW2800	Apartment / Unit / Flat	2	1	1			Sold by private treaty 10 Sep 2025 Sc	10/09/2025	Apartments		Terrace
$590.000	13/26 Warrendine Street,	ORANGENSW2800	Apartment / Unit / Flat	2	1	1			Sold by private treaty 08 Sep 2025 Sc	08/09/2025	Apartments		Villa
$460.000	4/190 McLachlan Street,	ORANGENSW2800	Apartment / Unit / Flat	2	1	1	188	m2	Sold by private treaty 05 Sep 2025 Lu	05/09/2025	Apartments		
$568.000	2/54a Amana Circuit,	ORANGENSW2800	Apartment / Unit / Flat	2	1	1			Sold by private treaty 22 Aug 2025	22/08/2025	Apartments		
$288.000	16/14-16 Burrendong Way	ORANGENSW2800	Apartment / Unit / Flat	2	1	1	89	m2	Sold by private treaty 01 Aug 2025 Ja	01/08/2025	Apartments		
$435.000	1/68 Margaret Street,	ORANGENSW2800	Apartment / Unit / Flat	2	1	1	208	m2	Sold by private treaty 31 Jul 2025 Jac	31/07/2025	Apartments		
$610.000	1/26a Cox Avenue,	ORANGENSW2800	Apartment / Unit / Flat	3	1	1			Sold by private treaty 18 Jul 2025 Scc	18/07/2025	Apartments		

median $687.500

Comparable figure (and reference):



The unit price in Orange (NSW) in Dec 2020 Nov 2021: $391,250

Source: realestate.com.au (2025). Orange NSW property market data.
https://www.realestate.com.au/nsw/orange-2800/

Brief statement: (generative AI is not allowed for the statement)

4 years ago, the price for a unit in Orange was around $391,000. While in recent records it is around $687,000. Over 4 years the price has increased more than 50%. At this point, it is a good opportunity to sell units.

Question 4: Investigating the relationship between bedrooms and other variables (10 marks)

a. **Generate** a pivot table that shows relationship between the number of ***bedrooms*** and the ***type*** of property. Additionally, show the ***group*** in the table. (2 marks)

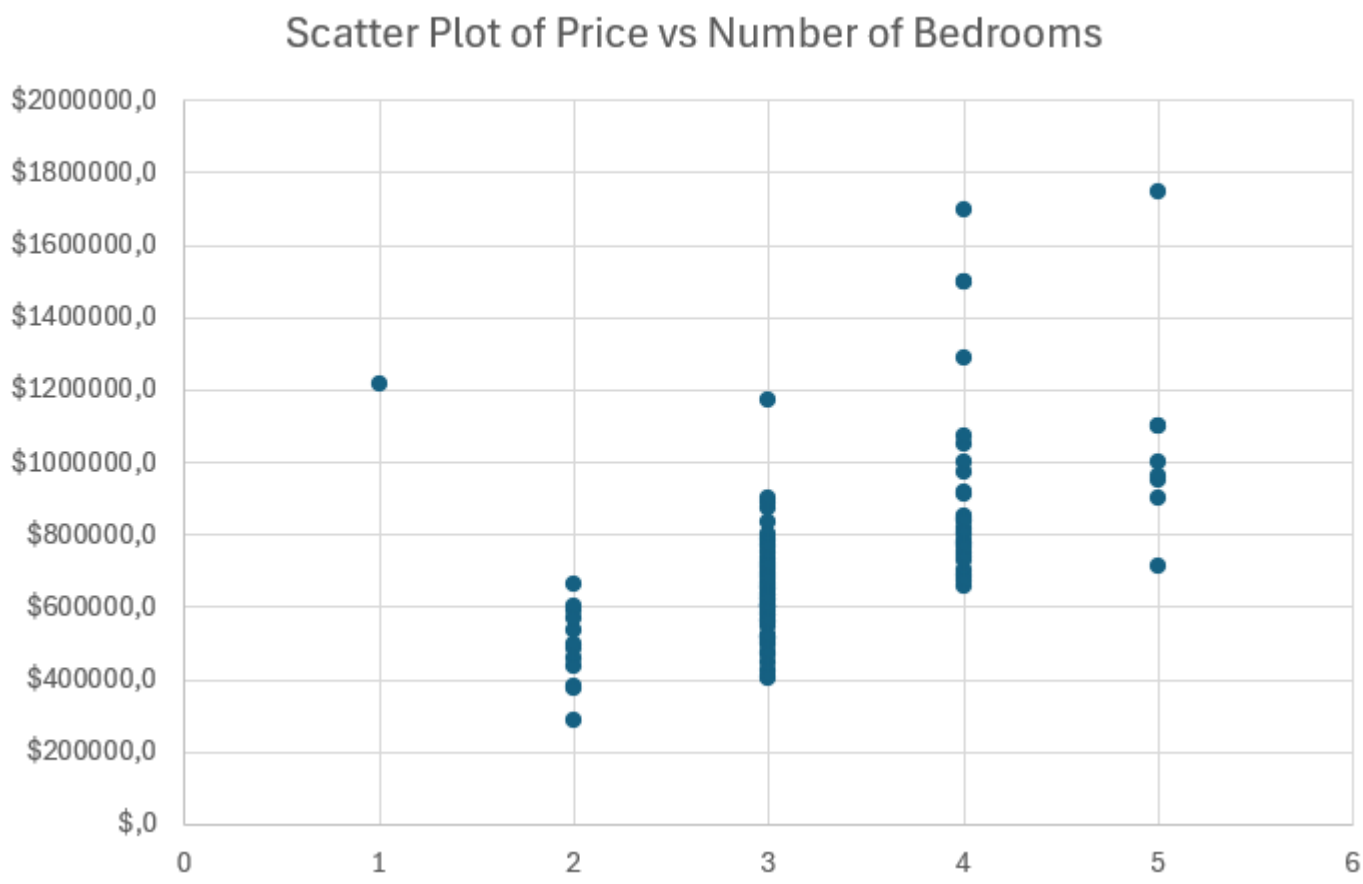
*Tips: use **type** and **group** (from Q2d) for the rows (what order is best?) and **beds** for the columns.*
Marking: 1 mark for showing counts of type vs beds, + 1 mark for incorporating group correctly.

Count of Type	Column Labels						
Row Labels	**1**	**2**	**3**	**4**	**5**	**(blank)**	**Grand Total**
Apartments		**6**	**1**				**7**
Apartment / Unit / Flat		6	1				7
houses	**1**	**6**	**56**	**29**	**8**	**1**	**101**
Block of units					1		1
House	1	6	52	29	7	1	96
Semi-detached			1				1
Terrace			1				1
Villa			2				2
(blank)							
(blank)							
Grand Total	**1**	**12**	**57**	**29**	**8**	**1**	**108**

b. **State** the most common number of bedrooms for <u>each</u> group (apartments and houses). (1 mark)
The most common number of bedrooms for Apartments is 2 bedrooms and for houses is 3 bedrooms.

c. Generate **a scatter plot** that shows the relationship between the number of ***bedrooms*** and the ***price***. Put bedrooms on the x-axis and price on the y-axis. Ensure your scatter plot includes a clear title and properly labelled axes. (3 marks)

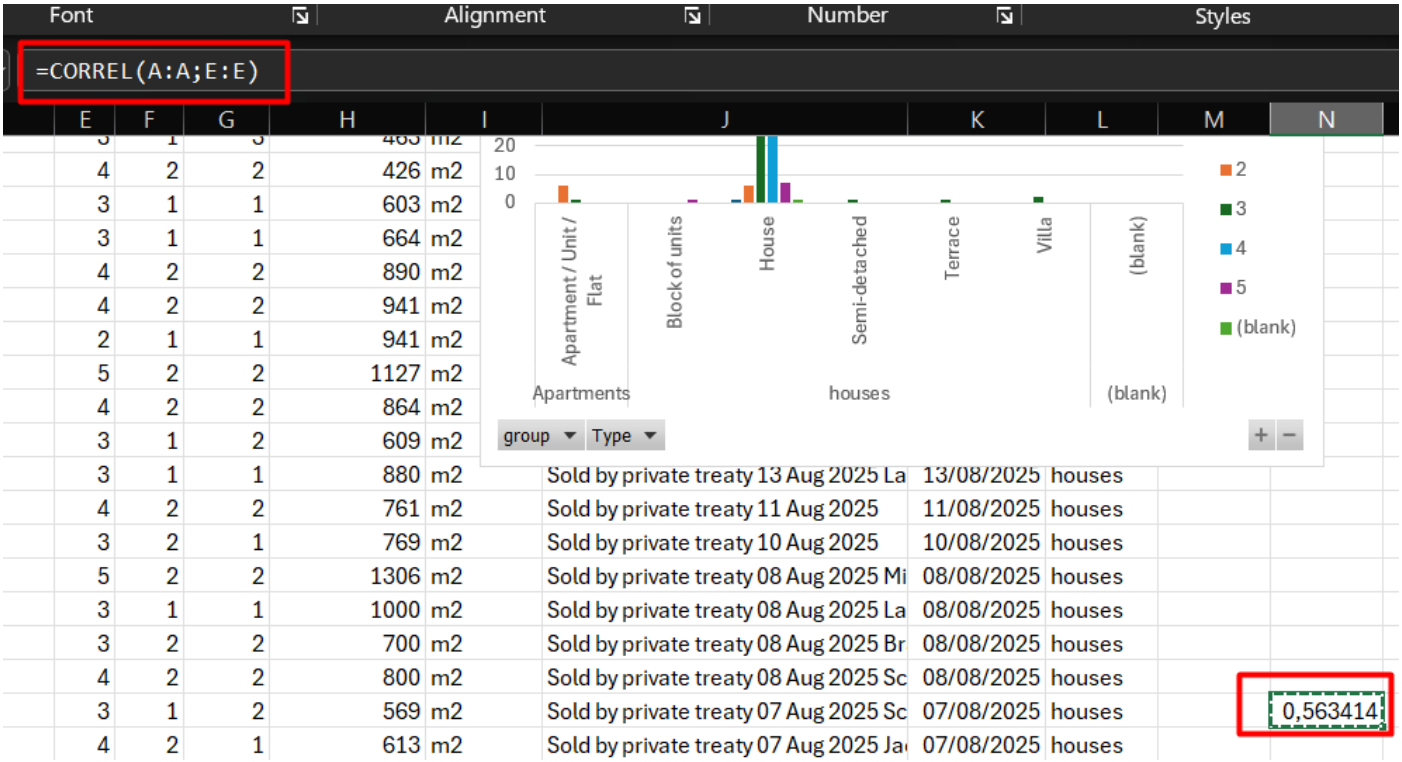
Tips: missing values can confuse Excel. So you are recommended to remove them (eg by filtering) before creating the chart. You looked at some missing values for Q1d.
Marking: 1 mark for scatter plot of beds and price, + 1 for title/axes/formatting, + 1 for beds on x and price on y.



d. **Find** the <u>correlation coefficient</u> between number of ***bedrooms*** and ***price***. **Show** how you used Excel to generate the number. (2 marks)

Correlation coefficient: 0,563414

Screenshot of Excel formula or menu used to generate:



>

e. Hence **describe** the relationship between price and the number of bedrooms. (2 marks)

The correlation between price and the number of bedrooms is 0.563414 what means a moderate correlation. While the chart shows that the more bedrooms, the price increases in correlation.

Question 5: Reflecting on the source of data (2 marks)

a. The master source of sales of properties in NSW is the NSW Land Registry Services. The data can be accessed from the NSW Valuer General site via https://valuation.property.nsw.gov.au/embed/propertySalesInformation. For your assignment though, we have sourced data from domain.com.au. **Discuss** the benefits and drawbacks of using the data from a commercial real estate website (like domain.com.au) instead of the NSW Valuer General site. (1+1 marks)

Tips: you could consider this question from a few different angles. One option is to look at the format of data from each (domain: https://www.domain.com.au/sold-listings/orange-nsw-2800/house/?excludepricewithheld=1) (Valuer General: https://www.nsw.gov.au/sites/default/files/noindex/2024-05/Property_Sales_Information_Data_Files_User_guide.pdf appendix 1/record B).

Answer here (Generative AI is not allowed for this question)
Benefits:

Using data from Domain is beneficial because it is more user-friendly and easier for non-expert users, which makes it better for users that aren't looking for much data.

Drawbacks:

However, a disadvantage of using Domain data instead of the NSW Valuer general data is that data may be incomplete and may not include all property sales, as it is designed for commercial purposes, it does not reflect the whole market.